Numericals on Module - I

Q.1.Find the built in barrier potential for silicon with the following data at room temp (27^0), N_A=$5x10^{15}/cm^3$, N_D=$2x10^{15}/cm^3$

Q.2.The built in barrier potential of GaAs at room temperature is 1.08 volt. Determine the intrinsic carrier concentration if concentration of donors and acceptor are $2x10^{15}/cm^3$ and $5x10^{15}/cm^3$

Q.3.A P^+N junction diode has an acceptor concentration of $10^{17}/ cm^3$

Find the built in barrier potential at room temperature if it is silicon.

Q.4. Germanium diode has a reverse saturation current in the diode when it is forward biased with 0.3 at 27^0C.

Q.5.The forward current in a silicon diode is 15 mA at 27^0C .If reverse saturation current is 0.24 mA .What is forward bias voltage? n=2 for silicon diode.

Q.6.A germanium diode carries a current of 10 ma when it is forward bias with 0.2 volt at 27^0C .Find (a) reverse saturation current (b)bias voltage required to get a current of 100 ma ,n=1 .

Q.7.A silicon diode has a saturation current is 3.2 mA what is the bias current when temperature changes to 102^0C.Take n=2

Q.8.Find the dynamic resistance of a PN junction diode at forward current of 2 mA. (Assume KT/q).

Q.9.Withref to the accompanying diagram find the dc resistance for diode at I_D= 2 mA, I_D= 2 mA,V_D= -10 V.

Q.10.When the emitter current of a transistor is changed by 1.1 mA, its collector current changes by 0.97 mA. Calculate (i) its common-base current gain.

Q.11.The dc current gain of a transistor in CE configuration is 89. Find its dc current gain in CC configuration.